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Co-design under the Bauhaus of the Seas Light-house Project: a New European Bauhaus case study in Lisbon and Oeiras

Cristiano Pedroso-Roussado
ITI / LARSyS, Instituto Superior Técnico – U. Lisbon, Portugal

Nicholas B. Torretta
ITI / LARSyS, Instituto Superior Técnico – U. Lisbon, Portugal

Frederico Duarte
ITI / LARSyS, Instituto Superior Técnico – U. Lisbon, Portugal; Faculdade de Belas Artes – U. Lisbon, Portugal

Luisa Seixas
ITI / LARSyS, Instituto Superior Técnico – U. Lisbon, Portugal; IHC, Faculdade de Ciências Sociais e Humanas – U.N. Lisbon, Portugal

Mariana Pestana
ITI / LARSyS, Instituto Superior Técnico – U. Lisbon, Portugal

See next page for additional authors

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Citation

Pedroso-Roussado, C., B. Torretta, N., Duarte, F., Seixas, L., Pestana, M., Labs Lda, Y., Lda, R., Light, A., and Nunes, N. (2024) Co-design under the Bauhaus of the Seas Light-house Project: a New European Bauhaus case study in Lisbon and Oeiras, in Gray, C., Ciliotta Chehade, E., Hekkert, P., Forlano, L., Ciuccarelli, P., Lloyd, P. (eds.), *DRS2024: Boston*, 23–28 June, Boston, USA. https://doi.org/10.21606/drs.2024.604

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Authors

Cristiano Pedroso-Roussado, Nicholas B. Torretta, Frederico Duarte, Luisa Seixas, Mariana Pestana, Yacooba Labs Lda, Redcatpig Lda, Ann Light, and Nuno Nunes

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DRS BOSTON 2024 JUNE 24 — 28

Co-design under the Bauhaus of the Seas Lighthouse project: A New European Bauhaus case study in Lisbon and Oeiras

Cristiano Pedroso-Roussado[a*], Nicholas B. Torretta[a], Frederico Duarte[a,b], Luisa Metello Seixas[a,c], Mariana Pestana[a], Yacooba Labs, Lda[d], Redcatpig, Lda[e], Ann Light[f,g], Nuno Jardim Nunes[a]

[a] ITI / LARSyS, Instituto Superior Técnico – U. Lisbon, Portugal
[b] Faculdade de Belas Artes – U. Lisbon, Portugal
[c] IHC, Faculdade de Ciências Sociais e Humanas – U.N. Lisbon, Portugal
[d] yacoobalabs.com, Av. Arriaga 30, Funchal, Portugal
[e] redcatpig.com, Terinov, Largo de Belém, Angra do Heroísmo, Portugal
[f] Malmö University, Sweden
[g] University of Sussex, United Kingdom

*Corresponding e-mail: cristiano.roussado@tecnico.ulisboa.pt

doi.org/10.21606/drs.2024.604

Abstract: The Bauhaus of the Seas (BoS) is one of the Lighthouse projects of the New European Bauhaus initiative (NEB). The project promotes the application of the NEB values – sustainable, beautiful, together – to develop locally grounded Demonstrator Pilots and focus cities' attention on the future of the oceans. A co-design approach is being applied in a series of participatory sessions with stakeholders, including nature/ecosystem experts, cultural institutions, local authorities and civic organizations, aiming at a consolidated collaborative approach. In this report we present the results from the co-design processes under development in Oeiras and Lisbon – two coastal territories involved in the BoS. Our findings reveal that co-designing in a multi-stakeholder participatory process presents challenges, ranging from the discomfort of working in bottom-up decision-making settings to the difficulty of amplifying underrepresented voices, as well as the ethical, philosophical and practical challenges of involving other-than-human beings.

Keywords: Co-design; Participation; New European Bauhaus; Bauhaus of the Seas

1. Introduction

The global environmental crisis, stemming from an extractive attitude towards other beings and the misconception of a separation between humans and nature, urges deep changes (Grinspoon 2016; Lovelock 2000; Latour, 2017). In a journey of reconnection with nature and

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regenerating relations with other beings, design plays a key role in challenging the mainstream anthropocentric approaches that value humans above other species. With this recognition of more-than-humans, we seek transformed cultures that open up for new regenerative relations with the world around us, not merely behaviour change and better use of materials (Akama et al., 2020; Giaccardi and Redström 2020; Light, 2023).

A political approach in this direction is the New European Bauhaus initiative (NEB), which attempts to leverage the historical role of the Bauhaus in transforming design and construction towards the European Green Deal. Poised as a pillar of the European research agenda, the NEB brings to the new Bauhaus the values of sustainability and inclusion in attempting to promote a new aesthetics for the 21st century (http://tiny.cc/iv8dvz). In 2022, several other cross-territorial Lighthouses Projects were deployed across Europe to exercise the NEB, such as:

- The CULTUURCAMPUS Lighthouse project aims to transform the disadvantaged urban areas of Rotterdam South (the Netherlands) through a blend of education, research, policy, and culture, considering the lived experiences of local communities (http://tiny.cc/fv8dvz);
- The DESIRE (Designing the Irresistible Circular Society) Lighthouse project aims to tackle climate change, biodiversity loss and use of resources, while exploring alternative ways of transforming territories in Denmark, the Netherlands, Slovenia, Italy, and Latvia by respecting the values of inclusivity, circularity and reconciling cities with nature (https://shorturl.at/abfmr);
- The EHHUR (EYES HEARTS HANDS Urban Revolution) Lighthouse project seeks to tackle socio-economic and cultural challenges such as social segregation, energy poverty, and degradation of depopulated historical centres in seven European countries (Denmark, Greece, Belgium, Portugal, Turkey, Croatia, and Italy) by supporting cities and its residents in transforming the built environment (https://shorturl.at/awzTY).

The Bauhaus of the Seas (BoS) approach, which stems from a manifesto created in 2020 by several researchers, designers and architects from Portugal (https://shorturl.at/kqG23), proposes a continental mobilization to achieve sustainable, inclusive, and aesthetical ambitions around water bodies. Grounded on this approach, the Bauhaus of the Seas Lighthouse project (BoS) is piloting new relations in seven EU locations based on four typologies of water ecosystems: Lagoon in Venice (Italy), River in Lisbon and Oeiras (Portugal), Sea in Öresund (Sweden) and Genoa (Italy), and Ocean in Delta (Rotterdam) (https://bauhaus-seas.eu). BoS puts forward an approach where Pilot projects seek to integrate an inter-disciplinary, intergenerational, and inter-species agenda to demonstrate the feasibility of the NEB values – beautiful, sustainable, together – in relation to these water bodies. Pilot projects are interpreted as 'drops' (Table 1), which will cause 'ripples' at the city scale, eventually causing 'waves' at a wider scale – and affect culture change. Achieving this would be proof of the concept that coastal cities can achieve the United Nations Sustainable Development Goals

(SDGs). Recognizing the environment, reconciling with the sea, reconnecting communities, and renewing practices are possible ways to legitimize and expand an existing diverse range of knowledge.

Table 1 Glossary of terms that are being explored in the BoS and represent its narrative.

Term	**Definition**
Demonstrator Pilot	Local projects led by the co-design process. The activities of the project happen through a culture-led, participatory and highly innovative process that embraces the values of sustainability, inclusivity, and aesthetics.
Pilot Team	The decision-making team composed by an academic institution, a cultural organization, and the local authority anchored in the particular coastal territory.
Sea Forum	The intergenerational, intercultural, interdisciplinary, and interspecies operative group around the Pilot Team that ensures grounding and provides input to co-pan and co-evaluate the drops' activities and their ripples.
Drops, Ripples, Waves	Drop is the name given to the Demonstrator Pilot initiatives that respond to site-specific challenges and generate concrete activities and experiments that engage communities. Ripples are the effects of the drop activities, resulting in the transfer of ownership to the relevant local actors to sustain legacy. Waves are the ultimate expected result of the Demonstrator Pilot, demonstrating the territorial, ecological, and community transformation derived from the process. So drops cause ripples that together potentially cause waves.
Executive Plan	Commonly used term to define a project's plan. For BoS it includes: the concept of the Demonstrator Pilot; a direction for the drop activities; a plan for the Zoöp Model; the expected impact and ripples; an evaluation plan; a preliminary plan for the long-term strategy.
Ocean Ambassadors	The people and organizations that work with public-facing activities and citizen's engagement during delivery of the drops.
Expert Development Program	A learning process that introduces new ways of working with the arts, culture, and creativity, by connecting Demonstrator Pilots to other relevant projects, practitioners, and methodologies, as well as between each other, through a series of inspiring, actionable, and NEB-related activities.

The BoS pilots bring together academic, cultural and territorial institutions (i.e. local authorities) to build a co-design strategy that then connects with artistic and culturally inspired activities. The core of the co-design process is the Sea Forum (Table 1), which brings together a broad range of sustainability and local nature and ecosystems experts, cultural practitioners, local organizations and authorities, who are invited to participate in the pilot's decision-making process and to steer its focus and approach.

In this paper, we present, report and discuss the co-design process of the Sea Forums in Lisbon and Oeiras (Portugal), which co-developed the Executive Plans for the Demonstrator Pilots (Table 1) that will be developed in the project's next phase. The Lisbon and Oeiras Pilot Teams (Table 1) include the Instituto Superior Técnico (IST), part of Lisbon University, as leading academic partner; the Calouste Gulbenkian Foundation (CGF) as cultural partner, and the municipalities of Lisbon and Oeiras as territorial authorities. The Lisbon pilot comprises two culturally inspired demonstrators: the Regenerative Menu and the Blue Makerspace (https://shorturl.at/egr49). The Regenerative Menu promotes the implementation of foodstuffs and culinary proposals that have a regenerative purpose in the local aquatic ecosystem and are flexible in their design to adapt to future climate changes. The Blue Makerspace is intended to create a design research space and a knowledge-sharing program that focuses on water-based materials from a non-extractivist perspective. The Oeiras pilot focuses on an Ocean Literacy Program that seeks to "educate new generations to be more responsible and participative towards the sustainability of the oceans" (https://shorturl.at/egr49).

1.1 Co-design in NEB and BoS

The development process of the Lisbon and Oeiras pilots was based on a co-design process to allow the proper implementation of the NEB Compass[1]. Co-design in BoS is based on the commitment to co-design in NEB[2]. The European-wide NEB initiative started itself with a co-design phase, where a platform was created to gather ideas and contributions from anyone willing to join. The NEB compass is a framework that works on the NEB's core values – sustainable, beautiful, together – and puts them into practice through the following working principles:

- A participatory process: to consult, to co-develop, and to self-govern;
- A multi-level engagement: to work locally, to work across levels, to work globally;
- A transdisciplinary approach: to be multidisciplinary, to be interdisciplinary, to be beyond-disciplinary.

Co-design as a process focuses on bringing together various people to design together new solutions (Sanders and Stappers, 2008). Co-design thus opens the design process to include

[1] https://rb.gy/l1f8w
[2] https://rb.gy/pzh9a

non-designers. For this reason, a central concern of co-design has been exploring how to open up the process and include people from different professional areas (see for instance Lee, 2008; Sanders and Stappers, 2008; Pedersen, 2016). Additionally, co-design has been increasingly used as a core strategy where collaboration is promoted and takes the form of collective creativity (Sanders and Stappers, 2008). According to Cretney (2014) there is a need to articulate the resilience of local communities in design community-driven approaches to tackle environmental and social issues. The discourse of resiliency refers to how societies adapt to externally imposed changes (Joseph, 2013). Due to the current social and environmental challenges, cities face a resilience imperative (Lewis and Conaty, 2012; Baibarac and Petrescu, 2019). In this context, co-design emerges as a means to bridge the gap between a wide variety of stakeholders involved in life, as well as the design of cities. Despite the assumption that co-design needs to involve the end-users in the design process to gather a greater range of ideas and specific knowledge about the context (Sanders and Stappers, 2008; Lewis and Conaty 2012), the leap here attempted is that the co-design will foster the bottom-up decision-making process by including non-traditional decision-makers at the table, like misrepresented and underrepresented communities. Therefore, designers and non-designers will co-create throughout the process, disrupting the claim that the designer, the academic or cultural representatives, and the local authorities are the expert problem solvers. This happens through the designers' facilitation, where they address the participant's level of awareness about design choices while providing a range of information and knowledge about the consequences of alternative choices and asking for participants' opinions and contextual insights (Sui, 2003). In an era of increasing engagement between people about the decisions for the private and the shared collective good that may affect them, co-design offers an approach to mitigate the conflicts arising during such engagement in areas such as workplace design, human-computer interaction, and architecture and urban planning (Taffe, 2015). The challenge of implementing the co-design is increased when a diverse range of participants are involved, with no integrated framework or standard terminology (Taffe, 2015). However, this is the core of co-design as a practice rooted in participatory design, moving forward from user-centred design aiming at the democratization of innovation and design processes by emphasizing the role of the communities in decision-making processes (Gregory, 2003; Simonsen and Robertson, 2012).

The need for a (creative) participatory practice relates to the activation of empowerment processes that can strengthen the bonding throughout local networks and struggles (Hurtig, 2023). Making participation a routine practice has the potential to achieve more equitable configurations, giving space and relevance to different and new voices in the decision-making. All of this takes time, since the participation acknowledges a power transfer to 'weaker' groups. Additionally, co-design often follows a 'logic of war' due to the clashing interests of participants, where some may win, lose, or even retreat (Pedersen, 2016). Still, the role of a designer is more of a diplomat to design approaches that facilitate creativity and concessions, thus limiting concerns and barriers to participation. Involving non-designers in the co-design process may raise questions about the legitimacy and authority of the designer itself.

However, co-design methodologies using a diplomatic perspective call for the transcendence of the participants' positions and roles, allowing the expansion from "win, loose, or retreat" keeping the respect of the situated power structure of the group (Pedersen, 2016).

Known limitations and challenges of co-design are the lack of time to allow proper participation; frequent use of jargon increases the communication gap between experts, designers and lay people. Nevertheless, the interdisciplinary nature of co-design approaches and participation makes it necessary to compromise excellence within one discipline in order to achieve feasibility and success. However, the notion of success, failure, and levels of collaboration differs among participants and the ongoing and mandatory interdisciplinarity is situated in the particular local context (Cairns et al., 2020). So, participants must not be interpreted as stand-alone subjects, their participation extends beyond the physicality of the events and the occurring interaction, and there is no standard for what constitutes a good participation (e.g., silence can be a powerful form of participation) (Andersen et al., 2015).

Several projects have been employed to pilot at the local level the NEB principles. However, there is not enough data to assess their efficacy in achieving the multi-level objectives (Fondazione FORMIT, 2023). Still, preliminary data have been disseminated that elucidate the benefits, limitations, and opportunities of the co-design. In a recent report about six NEB projects (EHHUR, Urban ReLeaf, Rural Partners, eLabHauSE.rur, drOp, and NEB of the mountain), several actions were recommended to increase the robustness of the co-design approach, to facilitate the replicative effect of the NEB principles in other territories (Fondazione FORMIT, 2023), for instance:

- Develop specific indicators for the NEB Compass to ease the monitoring of how projects are achieving the NEB principles at the international, national, and local levels;
- Define in more concrete terms the aesthetic value;
- Program network events for peer-learning at the local, national, and EU level.

Local and regional authorities constitute a key partner in the implementation of NEB projects, and a reported lesson throughout several NEB projects is that more capacity building is needed to assure proper ownership and execution of the bottom-up and interdisciplinary projects.

Following such tradition, the BoS Lighthouse Project is also framed under a co-design framework. The BoS co-design process happens locally, involving key regional partners in each location. The Pilot Team guides and facilitates the work of the Sea Forum, composed of independent nature experts, local organizations and local practitioners. Together, the Pilot Team and the Sea Forum co-design the Executive Plan for every Demonstrator Pilot (Figure 1).

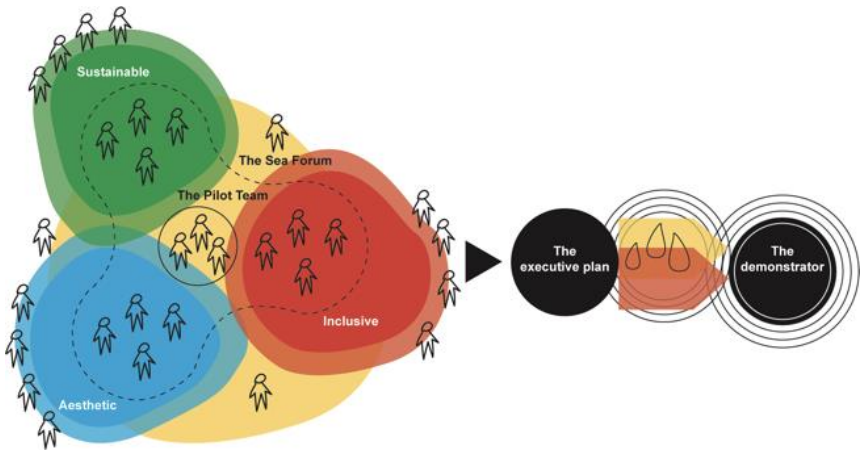


Figure 1 Diagram showing the relation between the BoS sliders, the Pilot Team, and the BoS co-design process towards the development of the Demonstrator Pilots Executive Plans (used with the permission of Anne Seravalli and colleagues – see BoS Deliverable 2.1[3]).

The BoS consortium put forward a co-design framework (see footnote 3) to be used by all pilot regions (see footnote 3). In this co-design template, BoS wanted to expand the co-design process to consider the Demonstrator Pilots' inter-disciplinary, inter-generational, and inter-species agenda. Therefore, throughout this agenda, a novel process of co-design emerges that is culturally driven, inclusive, respecting other-than-human beings, and locally grounded (Figure 2). To this end, values were drawn up to show where most communities might start in regarding their water ecosystem (on the left) and the BoS values (on the right) of relational aesthetics and an understanding of local and global interdependencies.

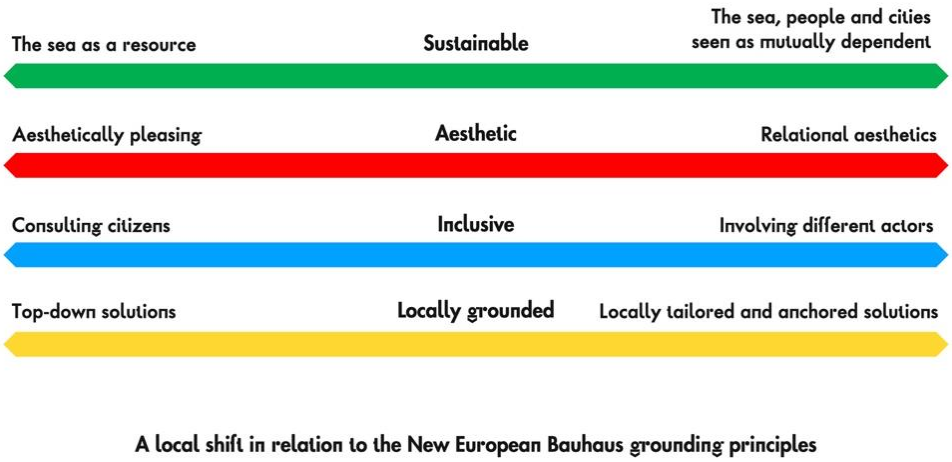


Figure 2 BoS sliders. A local shift to the NEB grounding principles. The aim is to walk the sliders from the left to the right position (used with the permission of Anna Seravalli and colleagues – see footnote 3).

[3] https://shorturl.at/ozHPX

The BoS co-design implementation was supported by two existing theoretical frameworks: "Theory of Change" (James, 2011; Taplin and Clark, 2012) and Zoöp (**https://shorturl.at/akyBI**). "Theory of Change" is a framework for developing projects with emphasis on the change needed and desired to bring the project to life. According to James (2011) "Theory of Change" is based on reflection about change, which opens and challenges how people see and understand the change needed for a specific project within a specific context and in collaborating with specific actors. The Zoöp is a framework for including non-humans in the decision-making process and regenerative impact of institutions that want to become a 'Zoöp'. According to the Nieuwe Institute, for BoS partners and proponents of the Zoöp, the framework allows for non-humans to influence processes immediately, without the need to wait for changes in laws and regulations.

2. Methods

The process implemented in the BoS Lighthouse Project for defining the local Demonstrator Pilots, replicated in the coastal locations, including Lisbon and Oeiras here analysed, starts from an intergenerational, intercultural, and interspecies co-design process. This co-design process integrates the diverse voices and interests of the communities (including the non-humans) and the underlying aquatic ecosystem. So, the co-design process enables the co-development of the local BoS vision of what quality climate-neutral transformation might mean in the local and ecological context – encompassing critical community systems and enabling conditions. In practical terms, it means that the BoS co-design process allows the exploration of possible matches for the needs of the local context with the existing possibilities from the Demonstrator Pilots.

2.1 Participants

The first step to select the participants was to form two Pilot Teams. They were composed of people affiliated with the IST - the academic partner -, CGF - the cultural partner -, and the Lisbon and Oeiras City Hall – as the local authorities, who sit one or more representatives in each Pilot Team. Then, a Pilot Team meeting was held to identify a list of potential Sea Forum participants whose profiles fit the NEB values. The profiles were: local nature experts (focusing on the Tagus River), local artists (regional or national), and representatives of local communities and organizations. Finally, the participants were recruited through an email sent by the academical partner to all the selected people. Throughout the co-design sessions the invitational effort was maintained to include the profiles that were being less represented.

2.2 Structure of the co-design process

For the co-design of the Executive Plans for the Demonstrator Pilots in Lisbon and Oeiras, eight Sea Forum events were organized from May to October 2023, four happening in each territory. The Lisbon Sea Forum's events were organized together for better facilitation. This

amplified the NEB values to include representatives drawn for their understanding of sustainability, inclusivity, aesthetics, and being locally grounded (see Figure 2).

The participants in each Sea Forum event were divided into groups of four to eight members. The teams were guided throughout the sessions to inform the Executive Plans for the Demonstrator Pilots. The Sea Forum co-design sessions followed a bottom-up approach guided by a common agreed vision for each Demonstrator Pilot, as described in the Introduction. The work followed a similar approach in Lisbon and Oeiras in accordance with the BoS narrative and workplan. Each of the co-design sessions were divided in four moments:

1. Overview of the BoS project: to increase the familiarity with its terminology and approach;
2. Summary of previous work: to keep participants updated with the state of the Demonstrator Pilot(s) Executive Plans;
3. Participatory work: to allow an open discussion and individual participation related to specific questions to be answered and added to the Executive Plans, and;
4. Wrap-up and final considerations: share insights from the session and the next steps.

Depending on the workload, a coffee-break occurred after the *Summary of previous work*, or in between the *Participatory work*. Since there were always new participants involved in the Sea Forum, keeping the *Overview of the BoS project* at the beginning was necessary, which also refreshed all the participants about the BoS terminology (Table 1) and narrative. Throughout the course of the Sea Forums events, the groups aimed to take ownership of the project and incrementally give feedback about the ongoing filling of the Executive Plans (see Table 2).

Table 2 Co-design methodology implemented throughout the sessions to facilitate the participation of the Sea Forum participants.

Activity	**First Session**	**Second Session**	**Third Session**	**Fourth Session**
Detailed concept of the Demonstrator Pilot	Q&A			
Detailed list of activities to be performed		Q&A	Q&A	
Zoöp-inspired plan				Zoöp
Expected impact of the of the activities and alignment with the local institutions and organizations	Q&A	Q&A		

Evaluation plan		Q&A	
Long-term strategy			"Theory of Change"

After the second session, three Executive Plans were drafted (two for the Lisbon Demonstrator Pilot, and one for the Oeiras Demonstrator Pilot) that included all the input the Sea Forum participants gave. These drafts aimed to share with the participants their contribution so far and help the Pilot Team tailor the following sessions regarding co-design tools and respective dynamics to apply.

The fourth session was divided into three parts to gather input from the Sea Forum participants about the long-term vision and objectives, and the impact on non-humans. Then, already in possession of the Executive Plan drafts, the Sea Forum participants were asked to apply a framework inspired by the "Theory of Change" (James, 2011; Taplin and Clark, 2012) and an adapted version of the Zoöp Model. The "Theory of Change" was used as inspiration to design a tool with the following steps[4]:

1. Identifying long-term goals;
2. Backwards mapping and connecting outcomes;
3. Completing the outcomes framework;
4. Identifying assumptions;
5. Developing indicators; and
6. Identifying interventions.

Specifically, the Sea Forum participants were asked to work on a simplified version of those six objectives. The session occurred as follows:

1. Define long-term goals - two per working group;
2. Define preconditions and requirements for those goals and briefly explain why - two per goal;
3. Define initial steps - one per precondition;
4. Define key-performance indicators (including a particular audience, a target, a threshold, and a timeline) - one per precondition.

Concerning the impact on non-humans, the Sea Forum participants were asked to immerse themselves in the other-than-humans potentially occurring in the respective areas. For that, the Zoöp Model was used as inspiration to guide the participants. The vision is to have all life

[4] https://rb.gy/woafg

forms cooperating, and it is inspired by the recent developments in the field of rights-of-nature implementation (Tola, 2018; Geddis and Ruru, 2019; Rosén et al., 2022; Crews, 2023; Light, 2023) with insights of the doughnut economy (Raworth, 2017). The Zoöp refers to the following steps:

1. Identifying – what bodies constitute the whole ecosystem where the Demonstrator Pilot will occur?;
2. Observing – what are the life-worlds of these bodies?
3. Characterizing – do these bodies form degenerative, regenerative, neutral relations?
4. Focusing – what are the pain points? Which should be our first regenerative goals?; and
5. Intervening – what actions should we plan to realize these goals?

In practice, the Zoöp model occurs as follows:

1. A Zoönomic Baseline assessment: composed of other-than-human beings, organizational, and legal entities or bodies, human artifacts and the built environment, among others;
2. Attribution of a Speaker for the Living: who is the person whose responsibility is to represent the other-than-humans in the decision-making process;
3. Definition and publication of a Zoönomic Year Plan;
4. Quarterly publication of the interventions and effects of the operation;
5. Publicly advertise the Zoöp manifest and the constitution of a Zoöp.

Therefore, the last exercise of the fourth session was to ask the Sea Forum participants to work on a simplified version of the s Zoöp model and framework. The exercise occurred as follows:

1. Define a Speaker for the Living and potential *Ocean Ambassadors* (i.e., a term used in the BoS terminology for people who can take the lead for the Demonstrator Pilot objectives in helping the cities' residents learn to care more for and about the sea, see Table 1) - two per working group;
2. Define five bodies to form the Zoöp that relate to the Demonstrator Pilot (biological/nature, social, built environment) - five per group;
3. Detail the relationship between those five bodies (e.g., support, obstruct, ignore) - one per interactive pair of bodies;
4. Define interventions to achieve regeneration and integrity in the relationships previously identified - one per relationship.

One important note is that the "Theory of Change" and Zoöp-related work needed to be adapted to increase the participation of the Sea Forum members. This happened due to an observation made in the first iteration of such work in Oeiras (see following sections). Therefore, the fourth session related to the Lisbon Demonstrator Pilots had less input in these points.

Lastly, the combined input from the four co-design sessions was assessed and discussed in a Pilot Team meeting to complete the respective Executive Plans (Figure 1). The BoS process was agreed in such a way that the local authorities and the cultural partner would take ownership of the Demonstrator Pilots' implementation after the Execution Plans were defined.

3. Results and Discussion

3.1 Participants

Altogether, a total of 32 participants were involved in the Lisbon Sea Forum and 24 in the Oeiras' (Table 3).

Table 3 Number of the Sea Forum participants in Lisbon and Oeiras according to the BoS profiles.

Territory	Local nature experts	Local artists	Local communities or organizations
Lisbon	6	6	20
Oeiras	4	3	17

In the Lisbon Sea Forum, no participant was enrolled in all the sessions, five participants were engaged in three sessions, 12 in two sessions, and 15 participated only in one. In the Oeiras Sea Forum two participants were involved in three sessions, eight in two, and 14 only participated in one session. No participant was involved in all the four sessions. All the profiles were represented in all sessions in both territories.

3.2 Co-design process

The BoS co-design process entails a bottom-up approach in the decision-making process of developing the Executive Plans associated with the Demonstrator Pilots. A wide range of participants were invited to the Sea Forum to better inform the Executive Plans. These participants were shown the global vision of the BoS, including all the Demonstrator Pilot typologies (https://shorturl.at/egr49), and introduced to the NEB compass (see footnote 1). Moreover, it was briefly explained what the goals and purposes of the Sea Forum were and what was the relationship between the Sea Forum, the Pilot Team, the Executive Plans, and the Demonstrator Pilots.

Due to the aspirational nature of the BoS, deeply embedded in the inter-disciplinary, intergenerational, and inter-species agenda of the upcoming Demonstrator Pilots, most of the

participants felt inspired and eager to participate. This was observed not only during the co-design sessions but also at informal moments (e.g., during coffee breaks, before and after the sessions, and through messages sent on social media channels). Examples of participants' feedback were: "Please, count me in for the next session and any other project you were involved in!", or "I am deeply inspired, see you in the next session!", or even "I already miss our brainstormings!".

Another observation was that the Sea Forum participants shared that the number of co-design sessions was few, and more sessions would benefit the overall objective for the completeness of the Executive Plans. This observation supports what has been found in other co-design studies, where the rush to have results do not allow much time for building trust. Therefore, the power structures between participants are kept, not allowing the development of a more robust confidence and trust between them (Pedersen, 2016; Cairns et al., 2020).

Despite the oscillation of the participants in the different sessions (Lisbon Sea Forum; first session n = 11, second session n = 18, third session n = 8, fourth session n = 12. Oeiras Sea Forum; first session n = 3, second session n = 16, third session = 7, fourth session n = 4), the engagement was high, and the information and opinions gathered were rich and able to populate all the necessary sections of the Executive Plans. However, as observed by the oscillations in Oeiras Sea Forum participants, it was difficult to secure a consistency of participants and most attendees were institutional, with little practice in participating in co-design processes or bottom-up approaches. The same observation was made in the Lisbon Sea Forum, but to a lesser extent. These observations might be explained by the demands on the academic partner, who had to manage the diversity of agendas of the participants while performing the co-design facilitation in a low number of sessions to complete the Executive Plans. That said, where there is little experience of co-design, there is always the need for extra time to conduct sessions that involve all parties and at the same time work strategically towards complex goals. Coming to this situation, the main learning is the usual one – that insufficient time (and recompense for time) was budgeted.

Nevertheless, we successfully engaged a wide realm of expertise by introducing people from various institutions connected to the Tagus River (and other local aquatic ecosystems) bringing different perspectives. Our effort in including such a variety of expertise (from science to art to policy) provided a body of knowledge that will foster the next steps for the Lisbon and Oeiras Demonstrator Pilots. Additionally, we observed that the participants' confidence increased throughout the sessions (Figure 3).



Figure 3 Co-design sessions showing different ways of engagement with the participants in the Sea Forum. (top left): Second session in Oeiras where a very top-down approach was necessary. (top right): Third session in Oeiras where the co-design occurred in the garden in smaller groups. (bottom left): Second session in Lisbon, hosting group table discussions. (bottom right): Last session in Lisbon, with a much more fluid interaction in pairs.

Participating in bottom-up processes allowed a better embrace of the discomfort and the face of the unknown, which contributed to a more excellent and denser discussion and input. To exemplify exciting insights related to each of the Demonstrator Pilots:

- Regenerative Menu: identification of local chefs and practitioners already working on the sustainability and regenerative mindset in their own projects; and identification of invasive and new species to assess their potential of being added in people's diets;
- Blue Makerspace: identification of local "makerspaces" that are highly interested in collaborating and helping the development of the Demonstration Pilot on the ground; and identification of waste that can be used as a resource for the creation of regenerative materials;

- Ocean Literacy Program: the experience of the Oeiras City Hall in related projects that potentiate the *sea appeal* in the territory.

One of the handicaps of our co-design process was the low participation of local community representatives, like the underrepresented communities, youngsters, and the elderly. The barriers to communication (academic terminology, high-level and cross-territorial EU projects, complex and novel methodologies, *etc.*) may be the explanation that impeded proper contact with these members of the community. Hurtig (2023) questions the benefits that disadvantaged neighborhoods would gain from city-led participatory practices. It is important to critically assess the co-design activities and the future of the co-designed project. Taking this into consideration, Hurtig (2023) analyzed the three-year (2017-2019) public art program Imagina Madrid (Madrid, Spain). The ambition was to invite participants from the city's peripheral areas to think and design the future of their residential neighborhoods. Similar to the BoS approach, the Imagina Madrid was also cultural led, involving the co-creation between citizens and artists. Throughout his analyses, Hurtig (2023) raises a relevant issue which is related to the simultaneously positive and negative impacts of participatory projects on urban regeneration: who are the true beneficiaries of such powerful "new civil imaginary"? This issue can be traced back to eight major challenges that were also identified by our work:

1. Participation does not happen immediately. **It takes time** to build trust and to design the appropriate platform for participation;
2. There are **tensions** that undermine either the creativity or the functionality of the project. For instance, the power structure formed by the co-design participants cannot be neglected (see Hurtig (2023) for observed discrepancies and unfavorable hierarchies between all the stakeholders - e.g., artists, designers, citizens, representatives of cultural organizations, local authorities);
3. To achieve greater transformation in the space they occur, projects must **stay and remain** after the participatory and co-design approaches;
4. To locally anchor the initiatives is not trivial so the need to mediate and **negotiate** a multitude of agendas is paramount;
5. There are **bureaucratic and legal issues** to make use of the public spaces which may reduce the benefits for the community;
6. A bottom-up approach implies **transfers of power**, and the city is not used to shifting the decision-making to citizens, artists, or experts;
7. Inclusivity must be assured by making the extra effort to **avoid drop-outs** and gradually falling participation from people coming from less favored social structures; and

8. It is evident that more **clarity** on the objectives, frameworks, and the details about the co-design process must occur and all participants would benefit from this irrespective of their background and social power.

In possession of these learnings, we intend to use the Sea Forum and the other mechanisms included in the BoS Lighthouse Project (e.g., Ocean Ambassadors and Experts Development Programs (Table 1)) to overcome these challenges. At the same time, the Demonstrator Pilots start to gain shape during 2024 and we expect an expanded inclusivity throughout the implementation phase. Nevertheless, local cultures and their power relationships and structures cannot be neglected (Light and Akama, 2018). Therefore, the Demonstrator Pilots will be deployed in a place, in a time, with a history (Merritt and Stolterman, 2012). Since participation alone does not ensure empowerment, the features of social status and identity must be properly addressed throughout the different mechanisms and co-design approaches to be applied in the Demonstrator Pilots to allow the accessible respect and interpretation of culture, intertwined with the respective power relationships without surrendering the rigor of co-design (Pedersen, 2016). Additionally, we chose not to constrain the participants by any means, developing a semi-structured co-design approach. As observed in White and colleagues (2023), semi-structured approaches provide more flexibility throughout the participation at the expense of focus. This means that while creativity was given more space to occur (i.e., increasing the chance of more disruptive and informed ideas to occur), there was the need, at times, to re-explain the BoS narrative and concepts, and the co-design methodologies at hand (including the Zoöp Model and Theory of Change). This balance was partially achieved and a way forward to improve the process may be to use a more structured co-design process at later stages and sessions, allowing a better understanding of expectations and goals of to be achieved. In our perspective, the tentative intergenerational, interspecies, and intercultural nature of the Sea Forum may be sufficient to permit creativity to happen despite the use of more structured approaches.

As the participants were eager to know more about the Zoöp Model and to see how it could fit into the objectives and activities of the Demonstrator Pilots, we dedicated the last co-design session to this work. We included the work dedicated to the “Theory of Change” in the same session to have a better notion of how the non-humans could be represented in the long-term impact of the Demonstrator Pilots. Compared to the Q&A co-design sessions, the fourth session was less amenable. However, contrary to our expectations, the sessions where the Zoöp concepts were approached were well received and discussed by the participants. However, “Theory of Change” was difficult to explain, work on, and discuss. This might be related to the fact that the participants were mostly used to developing projects in short or mid-term longevity, in a top-down fashion, starting from the practical actions and questions and not by the long-term vision and objectives as “Theory of Change” presumes. Additionally, these difficulties were somehow expected since we employed a semi-structured co-design process, which gives more flexibility and ownership at expense of compre-

hension (White et al., 2023). Despite being less time- consuming, we had to put effort in facilitating and guiding the sessions, maybe because of the novelty of the NEB and BoS narrative and concept (White et al 2016).

Moreover, discussing the impact of activities on non-humans is "easier said than done" and more work must be done to accommodate a full assessment. For that purpose, BoS will guarantee a training program and further co-design sessions with the original promoters of the Zoöp Model – led by Klaas Kuitenbrouwer, affiliated with the Niewe Instituut (Rotterdam, the Netherlands), and director of the Zoönomic Institute (https://shorturl.at/pGHQT).

4. Conclusion

As we highlighted, BoS has an ambitious, innovative, central, singular, co-design process planned for implementation across all Demonstrator Pilots and BoS locations. However, each territory deals with a different body of water and has a vastly different social, political and environmental context. Despite the success of the co-design as a whole and the center stage we gave to the people and their different expertise, the nuances of the contexts we encountered in Lisbon and Oeiras made it challenging to carry out the co-design as it was initially intended. Thus, we tailored a flexible framework respecting the Executive Plans' needs but keeping in mind the novelty of the BoS narrative and terminology and the expected discomfort of the invited participants. Throughout the sessions we had the chance to shift the discourse according to the participants' feedback in Lisbon and Oeiras, which increased the richness of the inputs gathered and the general comfort during the co-design sessions.

Despite our successful efforts in aligning the multistakeholder Sea Forums, we believe the co-design frameworks and dynamics need to be selected with caution to increase the inclusiveness and diversity of participants. Our local tailored co-design approach worked well for the purpose of the Sea Forum objectives – to inform the Executive Plans for the Demonstrator Pilots. Therefore, one first insight arising from our work is that a single and standard co-design framework cannot be sufficient to engage the range of participants necessary to respect the NEB values at the local level. To this end, our BoS codesign template stresses the importance of the local and the locally-determined, as an addition to the three NEB qualities of beautiful, sustainable and together.

Lastly, five important lessons were gathered from this co-design process. First, the co-design process as developed in the BoS Lighthouse Project motivated and inspired people to know more and upgrade their responsibility in local decision-making. Secondly, the bottom-up approach entails additional challenges to broader participation, but it becomes agreeable when people get to practice, i.e. when the co-design methodologies are applied several times. So, we envision that more people get involved in the future and that the engaging and motivating environment we encountered and fostered between May and October 2023 can work as 'drops', to form 'ripples' effects and produce 'waves' that disseminate a more inclusive and representative participation. Third, the co-design sessions were not enough to fully develop the Executive Plans promised as part of the deliverables at an early moment in the

project; as such, the BoS consortium through the Local Pilot Teams must keep consulting the Sea Forum during the deployment phase and start the implementation of the Ocean Ambassadors and Expert Development Program to increase the reach of the sustainable, culturally led, inclusive and locally-grounded narrative. This will also allow proper execution of the selected activities that have already been included in the documents. The fourth important point is to complete the analysis of the impact on non-humans and make sure that the theoretical promise materializes in practice. Fifth, further work must be done to increase the diversity of co-design participants, including those whose voices are historically underrepresented.

As a final message, we assume that the BoS co-design process with the Sea Forum is a first step towards concretization for the NEB vision. As this process is being implemented in several EU coastal cities with many different cultural traditions and social dilemmas, sharing results and knowledge gathering will be of tremendous value for Europeans and other coastal inhabitants. While respecting the cultural differences and anchoring these co-design processes with the local opportunities and challenges, the next steps are related to the evolution of the Sea Forum, and other community-led mechanisms, and the demonstration that pilot projects that are sustainable, inclusive, and beautiful can be achieved.

Acknowledgements: This research was funded by the Portuguese Recovery and Resilience Program (PRR), IAPMEI/ANI/FCT under Agenda C645022399-00000057 (eGames-Lab). This article was partially supported by the Horizon Europe BIG (GA 952226), and Bauhaus of the Seas (GA 101079995) projects. ITI / LARSyS is a collaborative laboratory funded by the Fundação para a Ciência e Tecnologia (UIDB/50009/2020). The funders had no role in the study design, data collection and interpretation, or the decision to submit the work for publication. The authors want to thank Oeiras City Hall and Lisbon City Hall, and Calouste Gulbenkian Foundation for the support during the Sea Forum sessions. The authors acknowledge all the Sea Forum participants for their relevant contribution during the sessions. Lastly, the authors thank Anne Seravalli and colleagues for the permission to use the figures 1 and 2.

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About the Authors:

Cristiano Pedroso-Roussado is a Portuguese multidisciplinary researcher. He performs in three main domains: science (biology), creativity (design), and management (strategy). Currently he is a posdoc for the BoS project in ITI/LARSyS, studying how to increase the representativeness of other-than-humans in decision-making.

Nicholas B. Torretta is a Brazilian mixed-race artist with a trajectory focused on design decolonisation and pluriversality, trained in Brazil and Scandinavia. His research interests focus decolonial and counter-culture practices. Nicholas is a posdoc at the BoS project at ITI/LARSyS.

Frederico Duarte is a Portuguese design critic. He assumes his country's subaltern condition to curate, research and work. Frederico is an Assistant Guest Lecturer at the Faculty of Fine Arts (Lisbon), as a posdoc at ITI/LARSyS, and as the BoS Communication Coordinator.

Luisa Metello Seixas is an author, researcher, manager, wife, and mother. Her educational background is in design and multimedia arts, as well as studies in aesthetics and contemporary history. She currently manages the BoS project, bringing an interdisciplinary vision to collaboration.

Mariana Pestana is a Portuguese architect with a design practice focused on participation and a curatorial practice focused on futures and more-than-humans. She works as Assistant Professor at IST and as the Coordinator of Arts and Culture of the BoS project.

Yacooba Labs, Lda is a Portuguese company focused on developing innovative, intuitive products and services to make interacting Web3 easier for everyone. Yacooba is a member of the eGames Lab – a unique egames cluster for the development of creative industries' in Portugal.

Redcatpig, Lda is a Portuguese video game development studio for several platforms and gamifications solutions. Redcatpig is a member of the eGames Lab – a unique egames cluster for the development of creative industries' in Portugal.

Ann Light is an interaction theorist and design professor at University of Sussex, UK, and Malmö University, Sweden. She specializes in participatory practice, human-technology relations and collaborative future-making, focusing on the politics, ethics and agency of design and the more-than-humans

Nuno Jardim Nunes is a Portuguese Professor raised in Madeira Islands. Nuno brings perspectives from computer science and sustainable interaction design into scaling digital infrastructures and interventions. Currently, Nuno is the director of the Interactive Technologies Institute and coordinates the BoS project.